



Brief article

Conditioning task switching behavior



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ABSTRACT

Most reward studies focus on the reinforcement of simple tasks or stimulus-response rules. However, recent theories (re)emphasized that cognitive control representations should adhere to the same reinforcement learning principles as do more basic stimulus and response representations. This study focused on the act of switching between different tasks, and investigated the effects of disproportionately rewarding task alternations or repetitions in a cued task switching paradigm on subsequent voluntary task switching behavior (i.e., when participants could choose which task to perform). The results show that subjects who were more rewarded for task alternations (relative to those more rewarded for repetitions) showed more task switching behavior. Moreover, this increased task switching behavior also came with a cost, with participants more rewarded for task repetitions showing a better task focus (i.e., smaller task-rule congruency effects). These results demonstrate that reward can reinforce more abstract control representations, beyond low-level stimulus or response representations.

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1. Introduction

Imagine cooking dinner while keeping an eye on the kids, and suddenly the phone rings. A situation like this presents a daily challenge for our cognitive control abilities: we have to adapt our behavior in response to novel stimuli in the environment (i.e., the phone suddenly ringing), switch back and forth between multiple tasks (i.e., cooking, watching the kids, and picking up the phone), and regulate our habits and impulses appropriately. In distinguishing cognitive control from other functions of the brain, it is often pitted against more automatic forms of low-level learning, such as associative learning or reinforcement learning. According to this distinction, cognitive control is not learned, but, instead, originates from a general supervisory system that corrects low-level learned behavior. This way, cognitive control and (reinforcement) learning are often portrayed as opposite forces in driving our behavior – reminiscent of the century-old dichotomy between the works of Tolman (1925) and Thorndike (1911). However, inspired by computational models that effectively modelled control functions in terms of simple reinforcement learning algorithms (Botvinick, Braver, Barch, Carter, & Cohen, 2001; Holroyd & Coles, 2002; Verguts & Notebaert, 2008), recent theories started to argue against this dichotomy, claiming that cognitive control adheres to, and might even be embedded in, the same (reinforcement) learn-

ing principles that drive more “low-level” forms of learning (Abrahamse, Braem, Notebaert, & Verguts, 2016; Egner, 2014; Verbruggen, McLaren, & Chambers, 2014).

Central to reinforcement learning is the century-old *law of effect* which posits that learned behavior is more likely to re-occur when it is followed by a reward (Thorndike, 1911). A growing number of studies already started to investigate the impact of reward on cognitive control (for a recent review, see Botvinick & Braver, 2015). However, these studies mostly focused on the anticipation of reward using explicit block-wise manipulations (e.g., Locke & Braver, 2008), cues (e.g., Padmala & Pessoa, 2011), or task features (e.g., Krebs, Boehler, & Woldorff, 2010; Umemoto & Holroyd, 2015) that predict reward. Although these experiments on reward anticipation are highly interesting for other reasons, they do not directly investigate the reinforcing effect of reward signals. To investigate how we learn from reward history (e.g., can reward strengthen the behavior that led to it?), we need studies that test the impact of reward delivery on subsequent behavior (for a similar distinction, see Notebaert & Braem, 2016).

This study zoomed in on the process of task switching, which is considered an important cognitive control function that allows us to flexibly switch attention between different tasks (Kiesel et al., 2010; Vandierendonck, Liefoghe, & Verbruggen, 2010). Specifically, an important open question is whether not only task sets themselves (Schiffer, Muller, Yeung, & Waszak, 2014; Umemoto & Holroyd, 2015), but also the overarching cognitive control process of task switching can be subject to reward modulations. No study thus far examined under closely controlled lab conditions

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whether people can be conditioned into more, or less, switching between tasks. To this end, this study will disproportionately reward task alternations versus task repetitions, and test whether this can impact task switching behavior in a subsequent voluntary task switching phase (see method). The anticipated results would be inconsistent with the idea of cognitive control as a strategic supervisory system that is insensitive to more basic forms of reinforcement learning, and contradictory to the notion that reward promotes exploitative behavior only (e.g., Thorndike, 1911); but concordant with the hypothesis that cognitive control is sensitive to reinforcement by reward, just as any other type of behavior (Abrahamse et al., 2016; Verbruggen et al., 2014).

2. Method

Subjects performed a typical task switching study where their task was to categorize words on either their animacy or size (for a review, see Kiesel et al., 2010). They were indicated which task to perform during the first half of each block (cued task phase), and were free to choose a task during each second half (free task phase). Crucially, half of the subjects was rewarded more on task alternations, whereas the other half was rewarded more on task repetitions, during the cued task phase. We studied their task switching behavior during the free task phase, when no more reinforcements were distributed. By using a stimulus set to which both tasks could be applied we could also examine the task-rule congruency effect (see below), which can inform us about whether the ability to maintain task focus (and its shielding from the other task set) was also affected.

Participants. 49 students (range = 18–33 years, 28 women, 12 left-handed) took part in return for 10 €. The study was approved by the Ghent University Psychology and Educational Sciences Ethical Committee. The sample size was determined using sequential Bayes hypothesis testing by increasing the sample until a decisive Bayes factor smaller than 1/6 or larger than 6 was obtained (p. 12, Schönbrodt, Wagenmakers, Zehetleitner, & Perugini, 2017).

Stimuli and material. The task stimuli consisted of 320 high frequent words, which could all be evaluated on whether they were living or non-living, and whether they were smaller or larger than a basketball. Both dimensions were crossed orthogonally resulting in four lists of 80 words (i.e., living/small, living/large, non-living/small, and non-living/large; for more information, see [Supplementary Material](#)). Importantly, these 320 words were presented in a random order and ensured the presentation of a novel stimulus on each and every trial. The task cues consisted of the vowels A, E, I, O, or U, or the consonants V, F, L, Q, or C, to indicate one task or the other. A task cue would never reappear within three consecutive trials. During the second half of each block, task cues would be replaced by the symbol # to indicate a free choice of tasks. Task cues were presented slightly above, while the stimulus words were presented slightly below, the centre of the screen (5% of the distance between the centre and the upper or lower boundary of the screen, respectively). Feedback was centrally presented and consisted of the “+10” symbols for high reward, and “+01” for low reward. The Dutch word for false (i.e., “FOUT!”) was used as error feedback in the training block. All instructions and stimuli were presented in a regular Arial 24-point font, in white on a black background. The left and right response keys were the letters S and D for the task assigned to the left hand, and the letters K and L for the task assigned to the right hand, on a standard QWERTY keyboard.

Procedure. The task was to categorize the word according to its animacy or size, depending on whether the above-presented letter was a vowel or a consonant. Specifically, participants had to press left (right) when the word was smaller (larger) than a basketball,

with one hand, and to press left (right) when the word was non-living (living), with the other. This letter-type-to-task and task-to-response-hand assignment was counterbalanced across subjects, for each reward condition separately (see below). Participants were explained that living could refer to every living organism, including animals, trees, plants, fruits, or vegetables. This way, half of the words required a left or right button press on both tasks (i.e., both hands), whereas the other half required a different side response per task. These words were categorized as congruent and incongruent trials, respectively. Last, the participant was instructed that on some trials they would see the # symbol, instead of a vowel or consonant, upon which they should choose a task randomly, as if decided by flipping a coin (i.e., free choice trials).

The reward instructions indicated that they could win a (multi-media) store coupon of 50€ if they won the most points out of all participants. They were explained that they could win points on every trial for correct responses within the response deadline, but that the amount of points would be randomly determined (which was untrue, see below): sometimes this could be one point, sometimes ten points. Last, subjects were instructed that they could not win points on free choice trials, but that accurate performance and an honest attempt to choose tasks randomly was still important to remain in the competition. In other words, a failure to respond accurately on most free choice trials, or a failure to choose tasks randomly (for example, by constantly repeating one task), would exclude them from potentially winning the store coupon.

Importantly, and unknown to the participant, half of the participants were assigned to the switching group, whereas the other half was assigned to the repetition group. Participants in the switching group could receive ten points on 80% (and one point on the remaining 20%) of the trials where the current task was different from the preceding task (i.e., a task alternation), while they could receive ten points on only 20% (and one point on the remaining 80%) of the trials where the current task was similar to the preceding task (i.e., a task repetition). This reward manipulation was reversed in the repetition group, where repetitions were more highly rewarded.

After a first run of twelve practice trials (with task cues, no rewards, and four words that were not part of the above-mentioned lists), participants had to inform the experimenter if the task was still unclear, after which they were shown the instructions once more, and presented with four blocks of 80 trials. Each block consisted of 40 trials using task cues followed by 40 trials using free choice cues. The exact randomization is explained in the [Supplementary Material](#). The trial procedure is visualized in [Fig. 1](#). Each trial started with the presentation of two vertically aligned central fixation crosses for 500 ms, after which the task cue replaced the upper fixation cross and stayed on screen for 1000 ms. Next, the target word replaced the lower fixation cross, and both target word and task cue remained on screen until response, or the response deadline of five seconds. Following response or the response deadline, the screen turned blank for 500 ms, followed by a feedback presentation of another 500 ms (high or low reward following correct responses, blank screen following incorrect responses or response omission), and an inter-trial interval of 1000 ms. Free choice trials were identical to forced choice trials, except for the task cue always being a #, and the absence of feedback.

Questionnaires. After the experiment, participants filled in the BIS/BAS questionnaire (Carver & White, 1994) to assess their reward responsiveness. Analyses with this scale are reported in the [Supplementary Material](#). Finally, participants were asked either verbally, or by questionnaire (the last nine subjects), whether they noticed an imbalance in the reward schedule: “Did

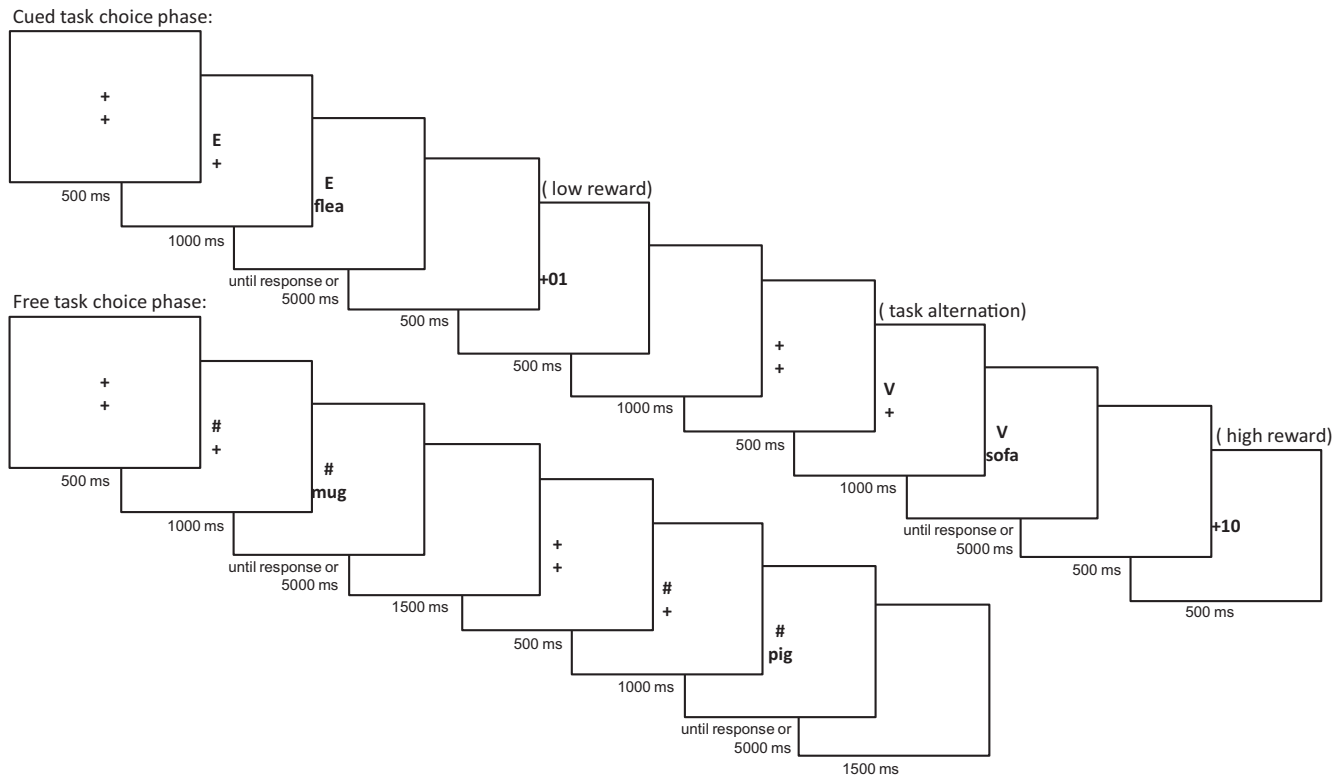


Fig. 1. General paradigm and trial procedure. Participants received 40 cued choice trials followed by 40 free choice trials in each of the four blocks. On each trial, the cue (consonant or vowel) indicated whether they had to categorize the word according to its animacy (living or non-living) or size (smaller or larger than a basketball). Number signs indicated a free choice of task. On cued choice trials, participants could also receive one or ten points following a correct response. Crucially, some participants received more points on task alternations, while others received more on task repetitions, thereby differently reinforcing task switching behavior.

you have the feeling that some type of trials were rewarded more (i.e., more +10's) than others, or did this seem completely random?". One participant from the switching group noted that task alternations were rewarded more, but results were the same with or without this participant.

Data analysis. Trials following an error or block transitions were removed from all analyses. The switch rate was calculated as the percentage of correct free choice trials that entailed a voluntary task alternation (relative to task repetitions), excluding the first trial of each free choice phase. Differences in switch rates between both groups were tested with both independent samples *t* tests and non-parametric Mann-Whitney *U* tests. Using the open source statistical program JASP Version 7.1 (Love et al., 2015), we also computed the Bayes factor BF_{10} for our main hypothesis (Cauchy prior width = 1, as specified by Rouder, Speckman, Sun, Morey, & Iverson, 2009), which quantifies the evidence for the alternative hypothesis against the null hypothesis. For reaction time (RT) analyses, only correct response times that were slower than 200 ms and within mean $RT \pm 2.5$ SDs were considered. Mean RT and error rates of the free choice trials were statistically analysed in repeated-measures analyses of variance (rANOVA) with the two within-subjects factors congruency (congruent versus incongruent) and task sequence (repetition versus alternation), and between-subjects factor reward group (switching versus repetition group). A similar analysis of the cued choice trials is reported in the [Supplementary Material](#).

3. Results

Seven participants were excluded from further analyses because they failed to perform an adequate number of task alternations on free choice trials (four of them belonged to the Switch

group; three to the Repetition group). Specifically, these participants alternated on 0.0–4.8% of the trials, possibly because they misread the instructions on the free choice task cues. Another four participants were removed from further analysis because their accuracy was more than two SDs below the mean (less than 76.2%). The remaining 38 participants (19 in each group) had a mean accuracy of 93.3% (SD = 4.8%) and alternated tasks on more than 12.5% of the trials. Some voluntary task switching studies (e.g., Arrington & Logan, 2005) used a more conservative threshold of 20% switch rate. Importantly, the present results reached the same levels of significance after removing two more participants who alternated tasks on less than 20% the trials (one in each group).

Switch rates Despite being instructed to choose tasks randomly, participants did show a bias (55.3%) towards choosing the size task, $t(37) = 3.076$, $p < 0.005$, but no significant bias (52.7%) towards choosing tasks assigned to the right hand, $t(37) = 1.424$, $p = 0.163$. Importantly, neither of those biases differed between groups, $t(36)$'s < 1 . Of main interest to our hypothesis was the voluntary task switching behavior (i.e., switch rate) on free choice trials. As hypothesized, the switching group alternated more between tasks than the repetition group (51.89% vs. 40.82%), $t(36) = 2.855$, $p = 0.007$, *Cohen's d* = 0.926, *CI* = [−18.9% −3.2%] (see Fig. 2). Specifically, the repetition group, $t(18) = -3.600$, $p < 0.005$, but not the switching group, $t(18) < 1$, showed a significant bias to repeat tasks (which is typically observed in voluntary task switching studies, Arrington & Logan, 2004, 2005; Orr, Carp, & Weissman, 2012; Vandierendonck et al., 2010). Interestingly, this group difference already reached significance on the very first block, $t(36) = 2.373$, $p = 0.023$, and remained significant for the three following blocks, all $t(36)$'s > 2.233 , $ps < 0.033$. This effect reached similar levels of significance with nonparametric *Mann-Whitney U* tests (overall

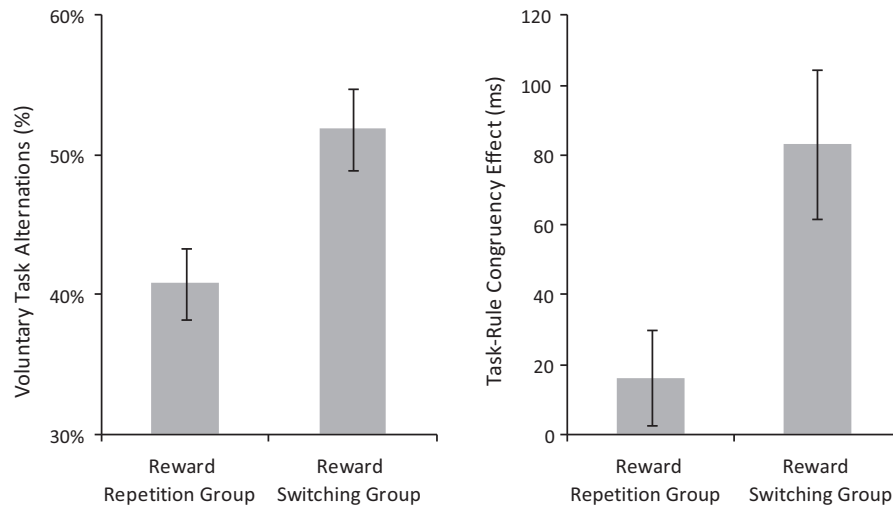


Fig. 2. The task alternation rates and task-rule congruency effects in the free task choice condition are depicted for each reward group separately. The group that was rewarded more for task alternations, relative to those rewarded more for task repetitions, showed more voluntary task switching behavior. Notably, this group also showed a larger task-rule congruency effect, suggesting that they were less efficient in shielding off the irrelevant task set. All error bars are ± 1 standard error of the mean (SEM).

$p = 0.005$; all p s per block < 0.043). The Bayes Factor BF_{10} for this group difference was 12.89, suggesting that these data are 12.89 more likely to be observed under the alternative hypothesis. This is considered strong decisive evidence for the alternative hypothesis that differentially rewarding task sequences can condition people into more or less voluntary task switching behavior (Jeffreys, 1961).

Reaction times and error rates. There was no reaction time difference in performing task alternations versus repetitions (i.e., task switch cost), $F(1, 36) < 1$, presumably because participants could choose the task order themselves and had sufficient time to prepare this (the combined inter-trial interval and fixation cross was 2000 ms). However, the RT analyses did show a significant congruency effect, $F(1, 36) = 15.14$, $p < 0.001$, as indicated by the slower reaction times on incongruent trials (1104 ms) versus congruent trials (1054 ms), indicating that the irrelevant task set interfered with performing the relevant task set. Most interestingly, this congruency effect significantly interacted with reward group, $F(1, 36) = 6.85$, $p = 0.013$, with the repetition group showing a smaller, non-significant, congruency effect (16 ms), $t(18) = 1.176$, $p = 0.255$, than the switching group, where the congruency effect was significant (83 ms), $t(18) = 3.871$, $p < 0.005$ (see Fig. 1). This difference in congruency effect was not further modulated by task sequence, $F(1, 36) < 1$. No other interactions were observed, all $F(1, 36)s < 1.9$, $p > 0.175$. The error rates also showed a significant congruency effect, $F(1, 36) = 4.81$, $p = 0.035$, but no task switch cost, or any interaction, all $F(1, 36)s < 1$.

4. Discussion

Thus far, studies of reward and cognitive control mainly focused on the motivating effects of reward anticipation (Botvinick & Braver, 2015). While some cognitive control studies did examine the more transient after-effects of reward feedback (i.e., on the previous trial; Braem, Hickey, Duthoo, & Notebaert, 2014; Braem, Verguts, Roggeman, & Notebaert, 2012; Fröber & Dreisbach, 2016; Hickey, Chelazzi, & Theeuwes, 2010; Jiang & Xu, 2014), no study thus far investigated the effects of a proportional reward learning phase, let alone its impact on voluntary behavior. Therefore, it remains an open question to which extent control functions can also be conditioned by rewarding their application. This study tested whether we can condition people into showing more or less

voluntary task switching behavior by disproportionally rewarding task alternations versus repetitions. The results clearly show that people can be conditioned into showing more, or less, task switching behavior (on free choice trials), when successful performance on task alternation trials were rewarded more, or less (on cued choice trials).

Interestingly, conflict processing was also affected on the free choice trials. Namely, people who were rewarded more for task alternations (and exhibited more voluntary task switching) showed a larger congruency effect. This finding is in line with the idea that task switching often interferes with conflict processing (Brown, Reynolds, & Braver, 2007), and the two processes reflecting antagonistic functions of stability and flexibility (Cohen, McClure, & Yu, 2007), presumably because task switching requires us to relax the process of task shielding (Dreisbach & Wenke, 2011).

Similar to the present study, Umemoto and Holroyd (2015) examined the effect of reward on task switching, but did so by selectively rewarding one task, but not the other (and instructed participants accordingly). In doing so, these authors demonstrated that switching away from the rewarded task was proven more difficult than switching to the reward task. As argued by Umemoto and Holroyd, the rewards possibly acted to increase task control over the rewarded task, thereby increasing the cost of breaking that control when switching to the non-rewarded task. Here, both tasks were rewarded equally, but rewards varied as a function of task repetitions versus alternations. Therefore, Umemoto and Holroyd (2015) studied reinforcement at the level of task sets, whereas the present study investigated reinforcement of transitions between different task sets, a process which is often situated one step higher in a hierarchy of task control (Holroyd & McClure, 2015). In our study, the above-mentioned “breaking of task control during switching” can be thought of as a candidate process which was more reinforced in the group that was more rewarded for task alternations.

Another related literature on model-based learning also demonstrated reinforcement learning of more abstract relations between behavior and reward that goes beyond the reinforcement of stimulus-response associations. For example, using a two-step decision task, studies have demonstrated how we can learn to use a statistical model of our environment to reach a response option of which we just learned its reward value, even when this

exact sequence of choices was not reinforced before (e.g., Daw, Gershman, Seymour, Dayan, & Dolan, 2011). These studies show how we are able to form hierarchical task sets that help obtaining rewards, an ability that has been linked to control functions which serve the exploitation of task sets (e.g., conflict processing, Otto, Gershman, Markman, & Daw, 2013; Otto, Skatova, Madlon-Kay, & Daw, 2014). In contrast, the present study showed the reinforcement of a control function itself, namely task switching, and its persistence in the absence of reward, by selectively reinforcing the repetition versus switching between two different task sets.

These findings support the idea that cognitive control representations are subject to reinforcement learning, just as stimulus/response representations are (e.g., Abrahamse et al., 2016), and are the first to show that the act of task switching can be conditioned by disproportionately rewarding its successful application. Interesting avenues for further research would be to determine the generalizability of such conditioning effects (i.e., transfer to other tasks that measure exploitative versus explorative behavior?), investigate whether they can be simulated by current computational models of cognitive control (e.g., the hierarchical control model by Holroyd & McClure, 2015), and study their application in clinical contexts (e.g., in patients with OCD or depression that show problems with task switching, Meiran, Diamond, Toder, & Nemets, 2011).

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Appendix A. Supplementary material

All data, experiment files, and analysis scripts can be found on the open science framework: <http://dx.doi.org/10.1016/j.cognition.2017.05.037>. Details on (the randomization of) the stimulus list, as well as supplementary analyses can be found in the supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.cognition.2017.05.037>.

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